

EMPLOYEES PROJECT

Import Libraries

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
pd.set_option('display.max_columns',None)
```

```
plt.style.use('ggplot')
```

Load Data

```
df = pd.read_csv("employee_data.csv")
```

```
df.head()
```

	Employee_ID	First_Name	Last_Name	Gender	Age	Department	Designation	City	Experience_Years	Salary	Bonus	Perfor
0	100001	Naveen	Verma	Male	28	Admin	Engineer	Vizag	1	2197515	140773	
1	100002	Amit	Mehta	Male	26	Sales	Consultant	Mumbai	5	2482524	147179	
2	100003	Sneha	Sharma	Male	41	HR	Executive	Delhi	22	351322	68801	
3	100004	Kiran	Rao	Female	46	Support	Analyst	Kolkata	19	2006698	236072	
4	100005	Kiran	Singh	Female	39	IT	Manager	Hyderabad	14	1440949	110762	

```
df.tail()
```

	Employee_ID	First_Name	Last_Name	Gender	Age	Department	Designation	City	Experience_Years	Salary	Bonus	Pe
114995	214996	Ravi	Sharma	Male	31	Finance	Manager	Kolkata	16	1930797	282270	
114996	214997	Pooja	Yadav	Female	34	Support	Consultant	Kolkata	19	1662753	47002	
114997	214998	Anjali	Kumar	Female	36	Sales	Executive	Delhi	23	858412	123835	
114998	214999	Ravi	Verma	Female	57	Sales	Engineer	Bengaluru	5	1892638	78927	
114999	215000	Kiran	Gupta	Female	24	HR	Executive	Mumbai	9	618867	160787	

```
df.shape
```

(115000, 18)

```
df.columns
```

```
Index(['Employee_ID', 'First_Name', 'Last_Name', 'Gender', 'Age', 'Department',  
      'Designation', 'City', 'Experience_Years', 'Salary', 'Bonus',  
      'Performance_Rating', 'Joining_Year', 'Work_Mode', 'Education',  
      'Marital_Status', 'Manager_ID', 'Leave_Balance'],  
      dtype='object')
```

Data Understanding

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 115000 entries, 0 to 114999  
Data columns (total 18 columns):  
#   Column                                Non-Null Count  Dtype  
---  -
```

```
---
0 Employee_ID      115000 non-null int64
1 First_Name       115000 non-null object
2 Last_Name        115000 non-null object
3 Gender           115000 non-null object
4 Age              115000 non-null int64
5 Department       115000 non-null object
6 Designation      115000 non-null object
7 City             115000 non-null object
8 Experience_Years 115000 non-null int64
9 Salary           115000 non-null int64
10 Bonus           115000 non-null int64
11 Performance_Rating 115000 non-null int64
12 Joining_Year    115000 non-null int64
13 Work_Mode       115000 non-null object
14 Education        115000 non-null object
15 Marital_Status  115000 non-null object
16 Manager_ID      115000 non-null int64
17 Leave_Balance   115000 non-null int64
dtypes: int64(9), object(9)
memory usage: 15.8+ MB
```

```
df.describe()
```

	Employee_ID	Age	Experience_Years	Salary	Bonus	Performance_Rating	Joining_Year	Manager_I
count	115000.000000	115000.000000	115000.000000	1.150000e+05	115000.000000	115000.000000	115000.000000	115000.000000
mean	157500.500000	39.982722	11.982357	1.374277e+06	154943.174365	2.996957	2017.998191	5997.31033
std	33197.784816	11.253322	7.206186	6.498925e+05	83609.343325	1.413232	4.897392	576.20432
min	100001.000000	21.000000	0.000000	2.500060e+05	10008.000000	1.000000	2010.000000	5000.00000
25%	128750.750000	30.000000	6.000000	8.114400e+05	82738.500000	2.000000	2014.000000	5499.00000
50%	157500.500000	40.000000	12.000000	1.376693e+06	154741.500000	3.000000	2018.000000	5996.00000
75%	186250.250000	50.000000	18.000000	1.936247e+06	227387.250000	4.000000	2022.000000	6495.00000
max	215000.000000	59.000000	24.000000	2.499994e+06	299999.000000	5.000000	2026.000000	6999.00000

```
df.dtypes
```

	0
Employee_ID	int64
First_Name	object
Last_Name	object
Gender	object
Age	int64
Department	object
Designation	object
City	object
Experience_Years	int64
Salary	int64
Bonus	int64
Performance_Rating	int64
Joining_Year	int64
Work_Mode	object
Education	object
Marital_Status	object
Manager_ID	int64
Leave_Balance	int64

```
dtype: object
```

```
df.isnull().sum()
```

	0
Employee_ID	0
First_Name	0
Last_Name	0
Gender	0
Age	0
Department	0
Designation	0
City	0
Experience_Years	0
Salary	0
Bonus	0
Performance_Rating	0
Joining_Year	0
Work_Mode	0
Education	0
Marital_Status	0
Manager_ID	0
Leave_Balance	0

dtype: int64

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df.nunique()
```

	0
Employee_ID	115000
First_Name	10
Last_Name	10
Gender	2
Age	39
Department	8
Designation	6
City	8
Experience_Years	25
Salary	112096
Bonus	94732
Performance_Rating	5
Joining_Year	17
Work_Mode	3
Education	7
Marital_Status	2
Manager_ID	2000
Leave_Balance	31

dtype: int64

```
df.memory_usage()
```

	0
Index	132
Employee_ID	920000
First_Name	920000
Last_Name	920000
Gender	920000
Age	920000
Department	920000
Designation	920000
City	920000
Experience_Years	920000
Salary	920000
Bonus	920000
Performance_Rating	920000
Joining_Year	920000
Work_Mode	920000
Education	920000
Marital_Status	920000
Manager_ID	920000
Leave_Balance	920000

dtype: int64

```
df.sample(10)
```

	Employee_ID	First_Name	Last_Name	Gender	Age	Department	Designation	City	Experience_Years	Salary	Bonus	Per
81394	181395	Sneha	Rao	Female	53	Sales	Consultant	Mumbai	16	668319	95336	
105825	205826	Amit	Kumar	Female	31	HR	Consultant	Pune	9	2385833	266316	
44366	144367	Naveen	Yadav	Female	50	Marketing	Executive	Chennai	2	2208460	64467	
72803	172804	Amit	Mehta	Male	48	IT	Consultant	Vizag	17	1902843	267138	
42760	142761	Ravi	Rao	Female	22	HR	Executive	Vizag	17	1600007	166080	
81092	181093	Rahul	Gupta	Male	22	Sales	Consultant	Mumbai	23	1919651	255571	
20584	120585	Vikram	Yadav	Male	53	Marketing	Executive	Mumbai	23	1939398	31112	
96958	196959	Amit	Sharma	Male	57	Admin	Executive	Vizag	11	1887921	10868	
22602	122603	Kiran	Singh	Male	53	Admin	Engineer	Mumbai	24	2390094	45361	
83579	183580	Naveen	Mehta	Male	56	Marketing	Consultant	Mumbai	17	727313	85080	

```
df.describe(include='all')
```

	Employee_ID	First_Name	Last_Name	Gender	Age	Department	Designation	City	Experience_Years	Sal
count	115000.000000	115000	115000	115000	115000.000000	115000	115000	115000	115000.000000	1.150000e
unique	NaN	10	10	2	NaN	8	6	8	NaN	N
top	NaN	Amit	Mehta	Male	NaN	Marketing	Lead	Chennai	NaN	N
freq	NaN	11681	11673	57579	NaN	14481	19246	14596	NaN	N
mean	157500.500000	NaN	NaN	NaN	39.982722	NaN	NaN	NaN	11.982357	1.374277e
std	33197.784816	NaN	NaN	NaN	11.253322	NaN	NaN	NaN	7.206186	6.498925e
min	100001.000000	NaN	NaN	NaN	21.000000	NaN	NaN	NaN	0.000000	2.500060e
25%	128750.750000	NaN	NaN	NaN	30.000000	NaN	NaN	NaN	6.000000	8.114400e
50%	157500.500000	NaN	NaN	NaN	40.000000	NaN	NaN	NaN	12.000000	1.376693e
75%	186250.250000	NaN	NaN	NaN	50.000000	NaN	NaN	NaN	18.000000	1.936247e
max	215000.000000	NaN	NaN	NaN	59.000000	NaN	NaN	NaN	24.000000	2.499994e

df.corr(numeric_only=True)

	Employee_ID	Age	Experience_Years	Salary	Bonus	Performance_Rating	Joining_Year	Manager_ID	L
Employee_ID	1.000000	0.003183	-0.003166	0.007432	-0.003768	0.000351	-0.002304	-0.004413	
Age	0.003183	1.000000	0.000014	-0.000898	0.000387	0.000518	-0.002433	-0.004851	
Experience_Years	-0.003166	0.000014	1.000000	-0.001413	0.001402	-0.003843	-0.003359	-0.005907	
Salary	0.007432	-0.000898	-0.001413	1.000000	-0.000450	-0.002749	0.003549	0.001772	
Bonus	-0.003768	0.000387	0.001402	-0.000450	1.000000	-0.000232	-0.002163	-0.001519	
Performance_Rating	0.000351	0.000518	-0.003843	-0.002749	-0.000232	1.000000	-0.004786	-0.000271	
Joining_Year	-0.002304	-0.002433	-0.003359	0.003549	-0.002163	-0.004786	1.000000	0.001254	
Manager_ID	-0.004413	-0.004851	-0.005907	0.001772	-0.001519	-0.000271	0.001254	1.000000	
Leave_Balance	0.001044	0.001177	0.005993	-0.001496	0.007949	-0.003707	0.000143	0.001961	

NumPy Analysis

salary = np.array(df['Salary'])

np.mean(salary)

np.float64(1374276.535095652)

np.median(salary)

np.float64(1376693.0)

np.max(salary)

np.int64(2499994)

np.min(salary)

np.int64(250006)

np.std(salary)

np.float64(649889.7231230099)

np.var(salary)

np.float64(422356652220.9024)

```
np.percentile(salary,25)

np.float64(811440.0)
```

```
np.percentile(salary,50)

np.float64(1376693.0)
```

```
np.percentile(salary,75)

np.float64(1936247.0)
```

Employee Analysis

```
df['Department'].value_counts()
```

count	
Department	
Marketing	14481
IT	14472
Sales	14421
Support	14408
HR	14403
Operations	14336
Admin	14244
Finance	14235

dtype: int64

```
df['Gender'].value_counts()
```

count	
Gender	
Male	57579
Female	57421

dtype: int64

```
df.groupby('Department')['Salary'].mean()
```

Salary	
Department	
Admin	1.371229e+06
Finance	1.373007e+06
HR	1.376878e+06
IT	1.374506e+06
Marketing	1.374433e+06
Operations	1.365340e+06
Sales	1.376619e+06
Support	1.382104e+06

dtype: float64

```
df.groupby('City')['Salary'].mean()
```

Salary	
City	
Bengaluru	1.370436e+06
Chennai	1.364130e+06
Delhi	1.383365e+06
Hyderabad	1.362645e+06
Kolkata	1.378041e+06
Mumbai	1.378822e+06
Pune	1.376711e+06
Vizag	1.380131e+06

dtype: float64

```
df.groupby('Education')['Salary'].mean()
```

Salary	
Education	
B.Sc	1.371776e+06
B.Tech	1.387578e+06
BCA	1.374612e+06
M.Sc	1.374713e+06
M.Tech	1.368052e+06
MBA	1.369915e+06
MCA	1.373228e+06

dtype: float64

```
df.groupby('Work_Mode')['Salary'].mean()
```

Salary	
Work_Mode	
Hybrid	1.373551e+06
Office	1.374915e+06
Remote	1.374361e+06

dtype: float64

```
df.groupby('Department')['Bonus'].sum()
```

Bonus	
Department	
Admin	2195755753
Finance	2206737800
HR	2244504830
IT	2241751758
Marketing	2245165335
Operations	2217383738
Sales	2238482976
Support	2228682862

dtype: int64

```
df.groupby('Gender')['Salary'].sum()
```

```
Salary

Gender
Female 78832316822
Male 79209484714

dtype: int64
```

```
df.nlargest(10, 'Salary')
```

	Employee_ID	First_Name	Last_Name	Gender	Age	Department	Designation	City	Experience_Years	Salary	Bonus	P
	91787	191788	Anjali	Yadav	Female	33	IT	Consultant	Vizag	12	2499994	63587
	108692	208693	Ravi	Rao	Female	38	IT	Manager	Hyderabad	18	2499982	62199
	98385	198386	Kiran	Gupta	Female	49	Admin	Manager	Kolkata	7	2499945	205272
	22040	122041	Vikram	Patel	Female	33	Admin	Analyst	Hyderabad	17	2499930	132395
	15576	115577	Sneha	Kumar	Male	32	Support	Manager	Bengaluru	22	2499917	19896
	38951	138952	Pooja	Singh	Male	24	HR	Executive	Chennai	15	2499889	191822
	46224	146225	Naveen	Verma	Male	23	Marketing	Consultant	Pune	21	2499854	268370
	55401	155402	Pooja	Mehta	Female	43	Finance	Executive	Hyderabad	8	2499844	97862
	23078	123079	Sneha	Sharma	Female	38	Support	Analyst	Pune	2	2499807	103339
	4213	104214	Amit	Sharma	Female	32	Support	Lead	Pune	16	2499801	21626

```
df.nsmallest(10, 'Salary')
```

	Employee_ID	First_Name	Last_Name	Gender	Age	Department	Designation	City	Experience_Years	Salary	Bonus	Per
	40448	140449	Pooja	Mehta	Female	49	Marketing	Executive	Vizag	17	250006	193002
	80415	180416	Kiran	Yadav	Male	43	Admin	Executive	Delhi	20	250038	261207
	11075	111076	Sneha	Reddy	Male	52	Marketing	Lead	Mumbai	23	250070	88052
	79441	179442	Vikram	Kumar	Male	43	Support	Analyst	Hyderabad	1	250110	44636
	4113	104114	Ravi	Kumar	Female	41	Sales	Engineer	Bengaluru	3	250159	279716
	94895	194896	Amit	Gupta	Male	44	Admin	Lead	Hyderabad	19	250174	261368
	10277	110278	Priya	Verma	Female	25	IT	Analyst	Kolkata	13	250181	110969
	9071	109072	Naveen	Patel	Female	51	Finance	Lead	Kolkata	16	250191	256171
	39312	139313	Pooja	Reddy	Male	39	IT	Consultant	Vizag	13	250202	235369
	18957	118958	Anjali	Rao	Female	26	Sales	Engineer	Chennai	24	250302	248380

```
df['Salary'].idxmax()

91787
```

```
df['Salary'].idxmin()

40448
```

```
df['Experience_Years'].mean()

np.float64(11.982356521739131)
```

```
df['Age'].mean()

np.float64(39.98272173913043)
```

```
df['Performance_Rating'].mean()
```

```
np.float64(2.9969565217391305)
```

Filtering

```
df[df['Salary']>1000000]
```

	Employee_ID	First_Name	Last_Name	Gender	Age	Department	Designation	City	Experience_Years	Salary	Bonus	P
0	100001	Naveen	Verma	Male	28	Admin	Engineer	Vizag	1	2197515	140773	
1	100002	Amit	Mehta	Male	26	Sales	Consultant	Mumbai	5	2482524	147179	
3	100004	Kiran	Rao	Female	46	Support	Analyst	Kolkata	19	2006698	236072	
4	100005	Kiran	Singh	Female	39	IT	Manager	Hyderabad	14	1440949	110762	
13	100014	Amit	Verma	Male	31	HR	Engineer	Vizag	22	1021244	144296	
...	...	...	...	...	...	...	...	...	...	...	...	
114993	214994	Naveen	Patel	Female	37	Support	Analyst	Delhi	2	1533833	138294	
114994	214995	Vikram	Sharma	Male	42	Sales	Consultant	Vizag	8	1633510	77327	
114995	214996	Ravi	Sharma	Male	31	Finance	Manager	Kolkata	16	1930797	282270	
114996	214997	Pooja	Yadav	Female	34	Support	Consultant	Kolkata	19	1662753	47002	
114998	214999	Ravi	Verma	Female	57	Sales	Engineer	Bengaluru	5	1892638	78927	

76574 rows × 18 columns

```
df[df['Department']=='IT']
```

	Employee_ID	First_Name	Last_Name	Gender	Age	Department	Designation	City	Experience_Years	Salary	Bonus	P
4	100005	Kiran	Singh	Female	39	IT	Manager	Hyderabad	14	1440949	110762	
5	100006	Pooja	Reddy	Male	57	IT	Consultant	Pune	4	845480	182555	
24	100025	Amit	Patel	Female	40	IT	Engineer	Mumbai	22	1609732	286842	
26	100027	Kiran	Mehta	Female	46	IT	Manager	Hyderabad	20	2153734	235844	
28	100029	Vikram	Reddy	Female	33	IT	Lead	Chennai	6	529041	289975	
...	...	...	...	...	...	...	...	...	...	...	...	
114937	214938	Ravi	Verma	Female	57	IT	Engineer	Pune	2	617010	15191	
114941	214942	Rahul	Gupta	Female	41	IT	Engineer	Kolkata	11	2129088	15341	
114948	214949	Sneha	Singh	Female	27	IT	Manager	Pune	7	1677929	295840	
114953	214954	Amit	Rao	Male	35	IT	Analyst	Vizag	22	1707999	267246	
114961	214962	Amit	Reddy	Female	33	IT	Lead	Vizag	11	1166646	97525	

14472 rows × 18 columns

```
df[df['City']=='Hyderabad']
```

	Employee_ID	First_Name	Last_Name	Gender	Age	Department	Designation	City	Experience_Years	Salary	Bonus	P
	4	100005	Kiran	Singh	Female	39	IT	Manager	Hyderabad	14	1440949	110762
	10	100011	Vikram	Gupta	Male	51	HR	Manager	Hyderabad	12	402400	10799
	12	100013	Amit	Yadav	Male	36	Operations	Lead	Hyderabad	11	792409	143767
	16	100017	Vikram	Gupta	Male	47	Sales	Manager	Hyderabad	22	1514806	72563
	26	100027	Kiran	Mehta	Female	46	IT	Manager	Hyderabad	20	2153734	235844
	...	...	...	...	...	...	...	...	...	...	...	...
	114921	214922	Ravi	Yadav	Female	28	Finance	Lead	Hyderabad	13	2319448	50246
	114967	214968	Naveen	Reddy	Male	45	Finance	Consultant	Hyderabad	15	2068846	279605
	114972	214973	Anjali	Kumar	Female	48	Operations	Lead	Hyderabad	22	1191571	115554
	114985	214986	Priya	Rao	Female	25	Admin	Lead	Hyderabad	4	2229729	93934
	114989	214990	Naveen	Patel	Female	22	Admin	Manager	Hyderabad	19	2232989	294096

14297 rows × 18 columns

df[df['Experience_Years']>10]

	Employee_ID	First_Name	Last_Name	Gender	Age	Department	Designation	City	Experience_Years	Salary	Bonus	P
	2	100003	Sneha	Sharma	Male	41	HR	Executive	Delhi	22	351322	68801
	3	100004	Kiran	Rao	Female	46	Support	Analyst	Kolkata	19	2006698	236072
	4	100005	Kiran	Singh	Female	39	IT	Manager	Hyderabad	14	1440949	110762
	7	100008	Sneha	Patel	Female	57	Marketing	Lead	Chennai	24	491235	107332
	9	100010	Naveen	Sharma	Female	27	Marketing	Manager	Delhi	17	749770	12571
	...	...	...	...	...	...	...	...	...	...	...	...
	114987	214988	Kiran	Gupta	Male	59	Marketing	Engineer	Vizag	17	1805521	253700
	114989	214990	Naveen	Patel	Female	22	Admin	Manager	Hyderabad	19	2232989	294096
	114995	214996	Ravi	Sharma	Male	31	Finance	Manager	Kolkata	16	1930797	282270
	114996	214997	Pooja	Yadav	Female	34	Support	Consultant	Kolkata	19	1662753	47002
	114997	214998	Anjali	Kumar	Female	36	Sales	Executive	Delhi	23	858412	123835

64231 rows × 18 columns

df[df['Performance_Rating']==5]

	Employee_ID	First_Name	Last_Name	Gender	Age	Department	Designation	City	Experience_Years	Salary	Bonus	P
	2	100003	Sneha	Sharma	Male	41	HR	Executive	Delhi	22	351322	68801
	7	100008	Sneha	Patel	Female	57	Marketing	Lead	Chennai	24	491235	107332
	12	100013	Amit	Yadav	Male	36	Operations	Lead	Hyderabad	11	792409	143767
	16	100017	Vikram	Gupta	Male	47	Sales	Manager	Hyderabad	22	1514806	72563
	17	100018	Rahul	Reddy	Male	36	Sales	Lead	Bengaluru	22	896447	259989
	...	...	...	...	...	...	...	...	...	...	...	...
	114961	214962	Amit	Reddy	Female	33	IT	Lead	Vizag	11	1166646	97525
	114973	214974	Pooja	Mehta	Male	59	HR	Engineer	Pune	23	2054684	247423
	114974	214975	Rahul	Kumar	Female	38	HR	Lead	Bengaluru	13	2126020	257831
	114981	214982	Kiran	Mehta	Female	43	Sales	Lead	Mumbai	6	446631	294115
	114997	214998	Anjali	Kumar	Female	36	Sales	Executive	Delhi	23	858412	123835

22857 rows × 18 columns

```
df[(df['Salary']>1000000)&(df['Department']=='IT')]
```

	Employee_ID	First_Name	Last_Name	Gender	Age	Department	Designation	City	Experience_Years	Salary	Bonus	P
	4	100005	Kiran	Singh	Female	39	IT	Manager	Hyderabad	14	1440949	110762
	24	100025	Amit	Patel	Female	40	IT	Engineer	Mumbai	22	1609732	286842
	26	100027	Kiran	Mehta	Female	46	IT	Manager	Hyderabad	20	2153734	235844
	76	100077	Anjali	Singh	Male	26	IT	Manager	Bengaluru	13	1558962	131255
	79	100080	Sneha	Singh	Male	51	IT	Engineer	Kolkata	13	1128802	248203
	...	...	...	...	...	...	...	...	...	...	...	...
	114894	214895	Rahul	Reddy	Male	49	IT	Consultant	Pune	19	1827405	227461
	114941	214942	Rahul	Gupta	Female	41	IT	Engineer	Kolkata	11	2129088	15341
	114948	214949	Sneha	Singh	Female	27	IT	Manager	Pune	7	1677929	295840
	114953	214954	Amit	Rao	Male	35	IT	Analyst	Vizag	22	1707999	267246
	114961	214962	Amit	Reddy	Female	33	IT	Lead	Vizag	11	1166646	97525

9693 rows × 18 columns

```
df[(df['Age']>30)&(df['Experience_Years']>5)]
```

	Employee_ID	First_Name	Last_Name	Gender	Age	Department	Designation	City	Experience_Years	Salary	Bonus	P
	2	100003	Sneha	Sharma	Male	41	HR	Executive	Delhi	22	351322	68801
	3	100004	Kiran	Rao	Female	46	Support	Analyst	Kolkata	19	2006698	236072
	4	100005	Kiran	Singh	Female	39	IT	Manager	Hyderabad	14	1440949	110762
	7	100008	Sneha	Patel	Female	57	Marketing	Lead	Chennai	24	491235	107332
	10	100011	Vikram	Gupta	Male	51	HR	Manager	Hyderabad	12	402400	10799
	...	...	...	...	...	...	...	...	...	...	...	...
	114992	214993	Rahul	Sharma	Female	44	Operations	Executive	Bengaluru	7	488664	36851
	114994	214995	Vikram	Sharma	Male	42	Sales	Consultant	Vizag	8	1633510	77327
	114995	214996	Ravi	Sharma	Male	31	Finance	Manager	Kolkata	16	1930797	282270
	114996	214997	Pooja	Yadav	Female	34	Support	Consultant	Kolkata	19	1662753	47002
	114997	214998	Anjali	Kumar	Female	36	Sales	Executive	Delhi	23	858412	123835

64886 rows × 18 columns

```
df[df['Gender']=='Female']
```

	Employee_ID	First_Name	Last_Name	Gender	Age	Department	Designation	City	Experience_Years	Salary	Bonus	P
	3	100004	Kiran	Rao	Female	46	Support	Analyst	Kolkata	19	2006698	236072
	4	100005	Kiran	Singh	Female	39	IT	Manager	Hyderabad	14	1440949	110762
	7	100008	Sneha	Patel	Female	57	Marketing	Lead	Chennai	24	491235	107332
	8	100009	Priya	Verma	Female	27	Sales	Executive	Mumbai	3	369657	116865
	9	100010	Naveen	Sharma	Female	27	Marketing	Manager	Delhi	17	749770	12571
	...	...	...	...	...	...	...	...	...	...	...	...
	114993	214994	Naveen	Patel	Female	37	Support	Analyst	Delhi	2	1533833	138294
	114996	214997	Pooja	Yadav	Female	34	Support	Consultant	Kolkata	19	1662753	47002
	114997	214998	Anjali	Kumar	Female	36	Sales	Executive	Delhi	23	858412	123835
	114998	214999	Ravi	Verma	Female	57	Sales	Engineer	Bengaluru	5	1892638	78927
	114999	215000	Kiran	Gupta	Female	24	HR	Executive	Mumbai	9	618867	160787

57421 rows × 18 columns

```
df[df['Work_Mode']=='Remote']
```

	Employee_ID	First_Name	Last_Name	Gender	Age	Department	Designation	City	Experience_Years	Salary	Bonus	P
	2	100003	Sneha	Sharma	Male	41	HR	Executive	Delhi	22	351322	68801
	4	100005	Kiran	Singh	Female	39	IT	Manager	Hyderabad	14	1440949	110762
	11	100012	Ravi	Verma	Female	57	Operations	Analyst	Mumbai	22	880862	209210
	14	100015	Amit	Verma	Female	42	Finance	Executive	Kolkata	15	2026656	206098
	16	100017	Vikram	Gupta	Male	47	Sales	Manager	Hyderabad	22	1514806	72563
...	...	...	...	...	...	...	...	...	...	...	...	...
114980	214981	Sneha	Yadav	Female	55	Finance	Manager	Vizag	22	2200552	208578	
114984	214985	Anjali	Patel	Male	51	Support	Engineer	Vizag	11	1883987	215031	
114988	214989	Amit	Sharma	Female	43	Sales	Consultant	Pune	5	1282451	184751	
114989	214990	Naveen	Patel	Female	22	Admin	Manager	Hyderabad	19	2232989	294096	
114996	214997	Pooja	Yadav	Female	34	Support	Consultant	Kolkata	19	1662753	47002	

38350 rows × 18 columns

```
df[df['Education']=='MBA']
```

	Employee_ID	First_Name	Last_Name	Gender	Age	Department	Designation	City	Experience_Years	Salary	Bonus	P
	7	100008	Sneha	Patel	Female	57	Marketing	Lead	Chennai	24	491235	107332
	12	100013	Amit	Yadav	Male	36	Operations	Lead	Hyderabad	11	792409	143767
	33	100034	Vikram	Singh	Female	31	HR	Consultant	Mumbai	21	555027	86464
	42	100043	Amit	Singh	Female	40	Admin	Lead	Delhi	13	375063	261473
	58	100059	Amit	Sharma	Female	41	Support	Lead	Delhi	9	1592923	241246
...	...	...	...	...	...	...	...	...	...	...	...	...
114962	214963	Ravi	Mehta	Male	32	Support	Manager	Vizag	18	1185905	51571	
114963	214964	Vikram	Verma	Female	32	Sales	Lead	Delhi	4	722193	235275	
114978	214979	Vikram	Reddy	Female	24	Finance	Manager	Bengaluru	19	524073	19131	
114989	214990	Naveen	Patel	Female	22	Admin	Manager	Hyderabad	19	2232989	294096	
114993	214994	Naveen	Patel	Female	37	Support	Analyst	Delhi	2	1533833	138294	

16241 rows × 18 columns

Feature Engineering

```
df['Total_Compensation']=df['Salary']+df['Bonus']
```

```
df['Salary_Lakh']=df['Salary']/100000
```

```
df['Age_Group']=pd.cut(df['Age'],bins=[20,30,40,50,60])
```

```
df['Experience_Level']=np.where(df['Experience_Years']>10,'Senior','Junior')
```

```
df['Bonus_Percentage']=df['Bonus']/df['Salary']*100
```

```
df['Salary_Rank']=df['Salary'].rank()
```

```
df['High_Paid']=np.where(df['Salary']>1000000,'Yes','No')
```

```
df['Age_Square']=np.square(df['Age'])
```

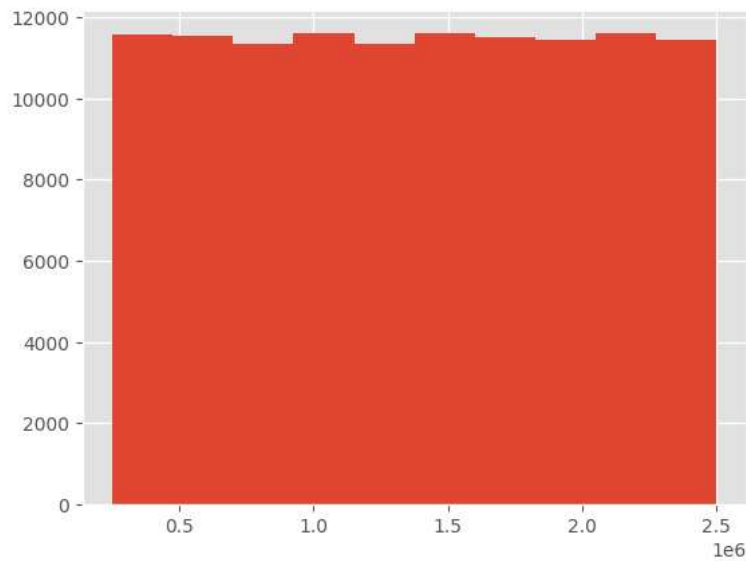
```
df['Log_Salary']=np.log(df['Salary'])
```

```
df['Tax']=df['Salary']*0.10
```

Matplotlib Charts

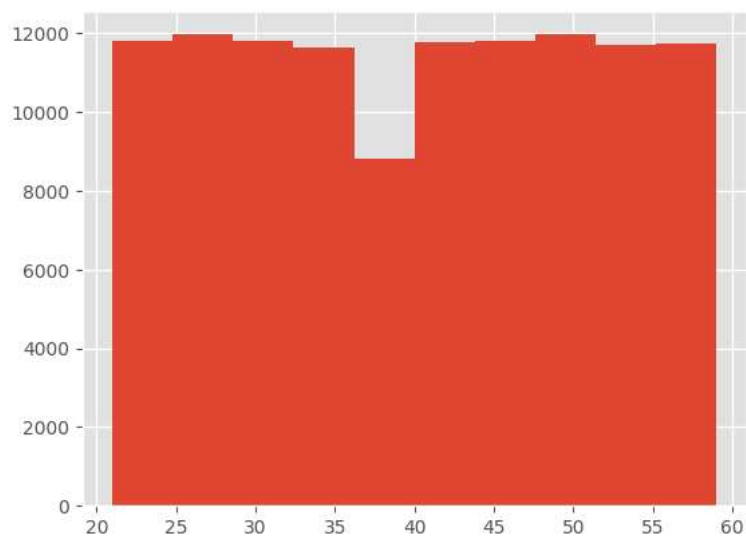
Salary Distribution

```
plt.hist(df['Salary'])  
plt.show()
```



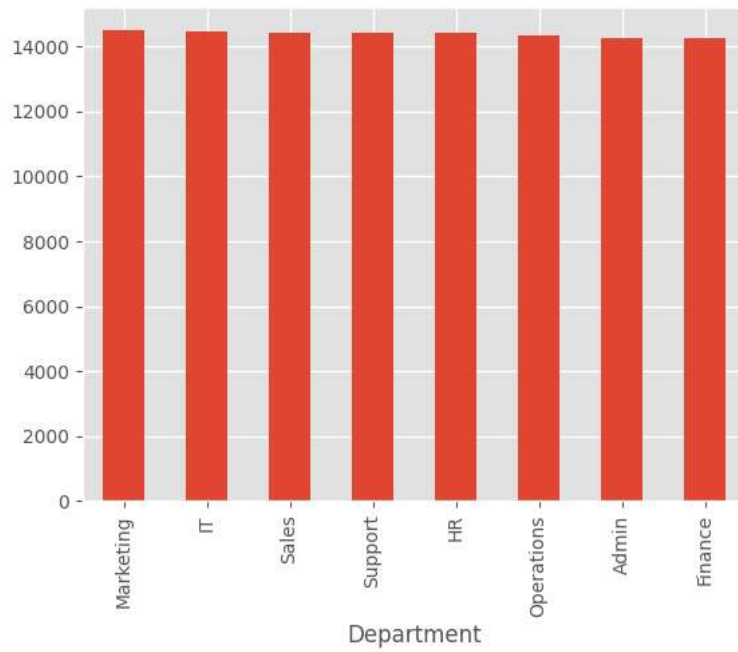
Age Distribution

```
plt.hist(df['Age'])  
plt.show()
```



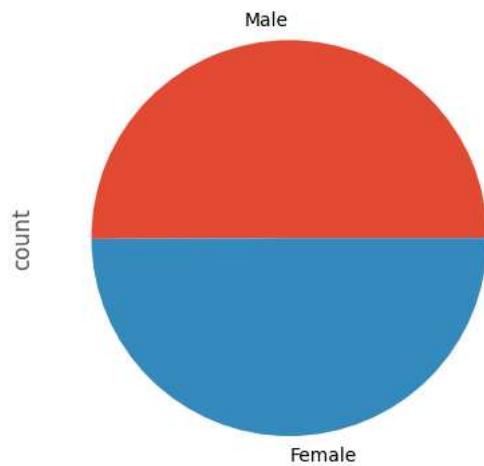
Department Count

```
df['Department'].value_counts().plot(kind='bar')  
plt.show()
```



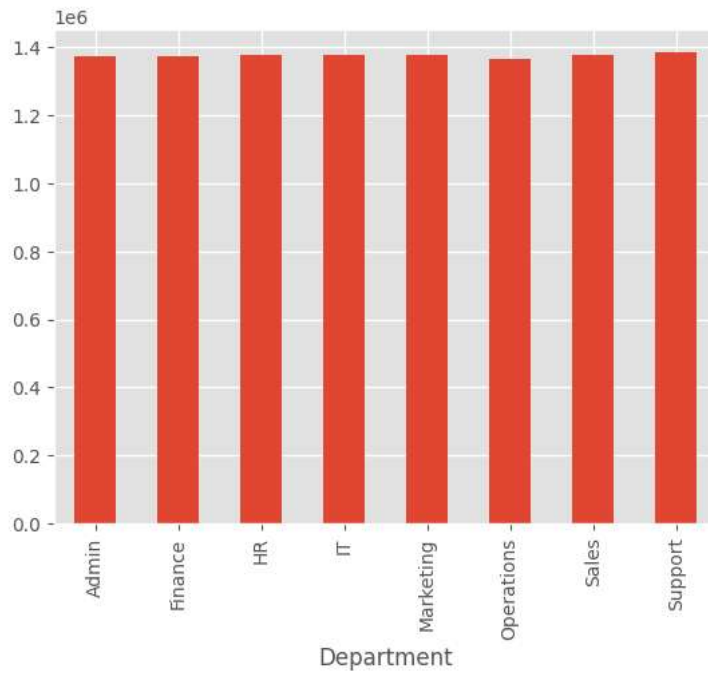
Gender Distribution

```
df['Gender'].value_counts().plot(kind='pie')  
plt.show()
```



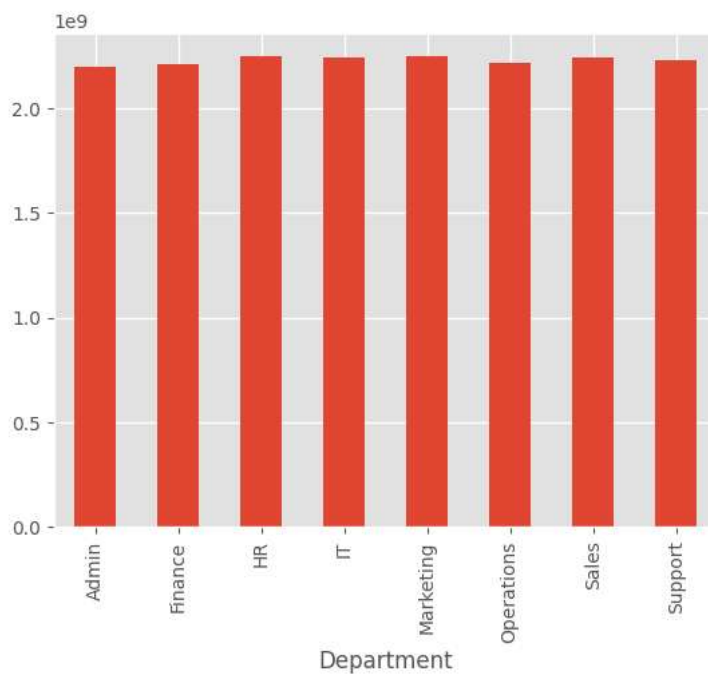
Salary by Department

```
df.groupby('Department')['Salary'].mean().plot(kind='bar')  
plt.show()
```



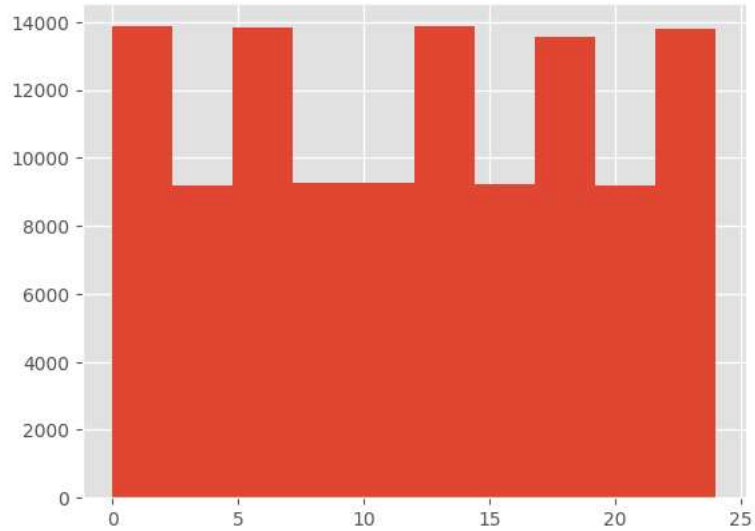
Bonus by Department

```
df.groupby('Department')['Bonus'].sum().plot(kind='bar')  
plt.show()
```



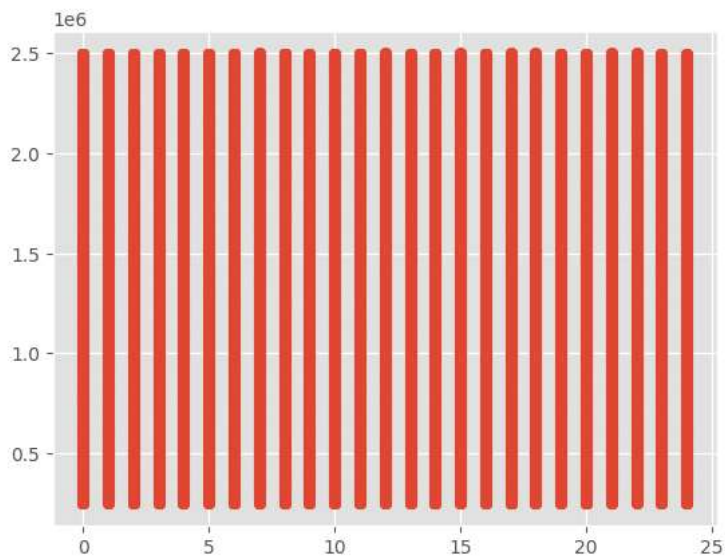
Experience Distribution

```
plt.hist(df['Experience_Years'])  
plt.show()
```



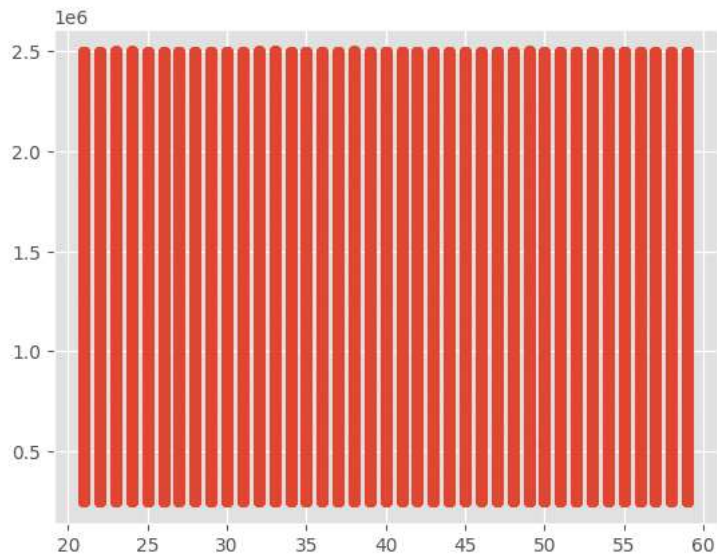
Salary vs Experience

```
plt.scatter(df['Experience_Years'],df['Salary'])  
plt.show()
```



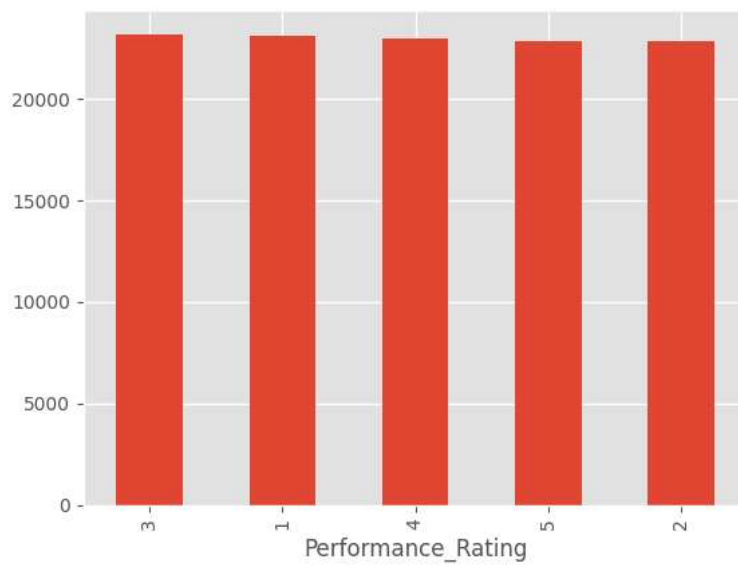
Age vs Salary

```
plt.scatter(df['Age'],df['Salary'])  
plt.show()
```



Performance Rating Count

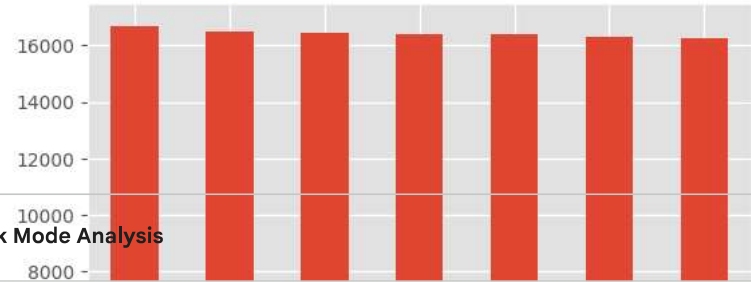
```
df['Performance_Rating'].value_counts().plot(kind='bar')  
plt.show()
```



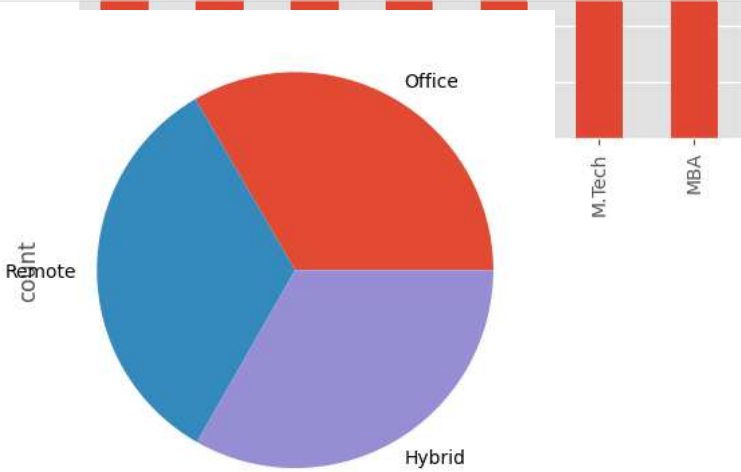
Education Analysis

```
df['Education'].value_counts().plot(kind='bar')  
plt.show()
```

Work Mode Analysis



```
df['Work_Mode'].value_counts().plot(kind='pie')
plt.show()
```



Salary Trend by Joining Year

```
df.groupby('Joining_Year')['Salary'].mean().plot()
plt.show()
```

